

Impatience is a virtue

CocoaHeads / SLUG June 2026

Saagar Jha

⌘ R

- ☒ Debugging MyApp on Saagar's iPhone 13 mini
- ☐ Copying OS library symbols from Saagar's iPhone 13 mini

dyld_shared_cache

```
(lldb) image list
```

dyld_shared_cache

```
(lldb) image list
```

```
[ 0] 9FE2065A-366A-3B37-85DD-5A71C5A600CF 0x0000000100000000 /System/Applications/Utilities/Terminal.app/Contents/MacOS/Terminal
[ 1] F15353E7-EABD-3B64-A3F8-3BA7A5923D46 0x0000000184f5b000 /usr/lib/dyld
[ 2] 07E8739B-6AEF-33AF-B71E-437F726569D5 0x00000001001dc000 /Users/saagarjha/Developer/dotfiles/libfixnano.dylib
[ 3] 61A5E926-E6F1-3E1F-AD69-9000FDFA3AAB 0x0000000100294000 /Users/saagarjha/Developer/dotfiles/stderrred/build/libstderrred.dylib
[ 4] 9C7A0538-AD9B-3E7F-96C0-2493153893A2 0x000000018bffa000 /System/Library/Frameworks/CoreGraphics.framework/Versions/A/CoreGraphics.framework
[ 5] 3BE2A7AA-7407-36BF-A473-FDA019BC0F07 0x0000000192981000 /System/Library/Frameworks/ImageIO.framework/Versions/A/ImageIO.framework
[ 6] 5AD26A6E-9D4C-386B-8373-2AE47610527F 0x0000000188334000 /System/Library/Frameworks/CoreText.framework/Versions/A/CoreText.framework
[ 7] 35775A6A-C24C-3290-9FDF-56900760DDA9 0x000000018c7a9000 /System/Library/Frameworks/ColorSync.framework/Versions/A/ColorSync.framework
[ 8] 6A94F8F4-88FB-3D3C-84EC-62421939FF3E 0x0000000198a79000 /System/Library/Frameworks/ApplicationServices.framework/Versions/A/ApplicationServices.framework
[ 9] 9FD71BB9-8BD2-3357-9966-21FCA8846A05 0x000000018c955000 /System/Library/Frameworks/ApplicationServices.framework/Versions/A/ApplicationServices.framework
[10] FEAACD75-F59D-37FD-AF9A-0CE26AAF6A48 0x00000001a461a000 /System/Library/Frameworks/ApplicationServices.framework/Versions/A/ApplicationServices.framework
[11] 8A15A5B5-0C70-3B44-942F-1D93000E6AC3 0x000000019a8bb000 /System/Library/Frameworks/ApplicationServices.framework/Versions/A/ApplicationServices.framework
[12] 248F701F-E33C-3776-AF5D-FA45065E6F27 0x000000019c9f5000 /System/Library/Frameworks/CoreServices.framework/Versions/A/CoreServices.framework
[13] 47533771-C07E-3375-A866-72B48176380A 0x000000019ac0c000 /System/Library/Frameworks/ApplicationServices.framework/Versions/A/ApplicationServices.framework
[14] 97E942EB-7E8F-3360-BD18-6ADE9C3E931C 0x0000000195e4b000 /usr/lib/libSystem.B.dylib
[15] F36943E2-71CB-38F4-82C2-6C44368D6BD4 0x000000019a8b0000 /System/Library/Frameworks/ApplicationServices.framework/Versions/A/ApplicationServices.framework
[16] 235DF1D1-0910-3C22-A2CE-76E8D2FF6F2F 0x000000018ca72000 /System/Library/PrivateFrameworks/ProtocolBuffer.framework/Versions/A/ProtocolBuffer.framework
[17] 566A00C9-7558-310A-9B5D-4DB75F20EECC 0x0000000195e4f000 /System/Library/PrivateFrameworks/SoftLinking.framework/Versions/A/SoftLinking.framework
[18] 63D913E3-D6AA-370E-8949-AC139799347E 0x000000018b9e0000 /System/Library/PrivateFrameworks/SkyLight.framework/Versions/A/SkyLight.framework
[19] 294AEC34-E798-3F2D-A727-0FBBA5AFCB34 0x000000019ac07000 /System/Library/Frameworks/ApplicationServices.framework/Versions/A/ApplicationServices.framework
[20] 863EA1B9-6891-362C-B23E-D6EB9A648472 0x00000001890e1000 /usr/lib/libicucore.A.dylib
[21] 4E0C3107-FC17-3FAB-A086-FA803AACD141 0x00000001923ab000 /System/Library/PrivateFrameworks/FontServices.framework/libFontParser.o
```

[232]	36A3179B-19D7-3773-9C09-17BDD27322B5	0x00000000203990000	/System/Library/PrivateFrameworks/MIL.framework/Versions/A/MIL
[233]	4B509E10-7698-3B2B-94D0-45E0DB48567A	0x00000000197b8d000	/System/Library/PrivateFrameworks/AppServerSupport.framework/Versions/A/
[234]	E7561FB8-4E88-3F15-98DF-1227C9759A30	0x000000002432b8000	/System/Library/PrivateFrameworks/Ecosystem.framework/Versions/A/Ecosyst
[235]	1832C524-133A-3C47-B1F4-F5F14B6B3525	0x00000000195de5000	/usr/lib/libcharset.1.dylib
[236]	D8D85C9C-2170-3656-9A97-1FFB51DB6449	0x00000000194807000	/System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/ve
[237]	5C343F67-0172-382E-B068-6ED28955E686	0x00000000211724000	/System/Library/Frameworks/OpenGL.framework/Versions/A/Libraries/libGLU.
[238]	67C36C10-8639-32AB-9117-5DC7C17D24E7	0x00000000195812000	/System/Library/PrivateFrameworks/AuthKit.framework/Versions/A/AuthKit
[239]	52021653-38AB-3036-ACB3-2155E2886EE8	0x0000000018d828000	/System/Library/Frameworks/Accounts.framework/Versions/A/Accounts
[240]	7673ED52-80A0-34D4-B1DF-C3360175BE0F	0x00000000195130000	/System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/ve
[241]	F0494AEA-AA21-3427-A131-6553F3CCE75A	0x000000001b5e56000	/usr/lib/swift/libswiftCoreImage.dylib
[242]	0C1252CA-40C1-32BC-B746-9BD0E09B6A9F	0x000000001bd144000	/System/Library/PrivateFrameworks/SoftwareUpdateCoreSupport.framework/Ve
[243]	E1014FD2-9E28-3254-AF06-C222DAC42ED9	0x000000001b7b51000	/usr/lib/libbootpolicy.dylib
[244]	47738005-72AB-30E9-B38C-39E77A89A945	0x000000001a5033000	/System/Library/PrivateFrameworks/StreamingZip.framework/Versions/A/Stre
[245]	C2F96FC7-3E5B-3532-839F-F57E55CBA0E0	0x0000000018affa000	/System/Library/Frameworks/UniformTypeIdentifiers.framework/Versions/A/U
[246]	0E805780-4CAE-30F8-ADF5-E8D5AFE66C5C	0x000000001a4269000	/System/Library/PrivateFrameworks/RemoteXPC.framework/Versions/A/RemoteX
[247]	0591DB1B-23AE-37DB-9C4D-4CB8A319E2A7	0x000000002116f0000	/System/Library/Frameworks/OpenGL.framework/Versions/A/Libraries/libGLIm
[248]	3BDA2590-D32C-36DA-9406-8CD6E21BBBFF	0x00000000258397000	/System/Library/PrivateFrameworks/InstalledContentLibrary.framework/Vers
[249]	24E1E562-C104-3D38-8AF4-B4C3D211E6CE	0x0000000019022c000	/System/Library/PrivateFrameworks/UserManagement.framework/Versions/A/Us
[250]	00E38C54-90FA-342D-B9C8-3E671FA28AF3	0x000000001a4974000	/System/Library/PrivateFrameworks/AppleIDSS0Authentication.framework/Ve
[251]	7CD3D8E9-753A-3F91-B399-B6435B774854	0x000000001bc879000	/System/Library/PrivateFrameworks/MSUDataAccessor.framework/Versions/A/M
[252]	E226BA2F-4B47-3528-A818-94530BFE0737	0x000000002aadb8000	/usr/lib/swift/libswiftSpatial.dylib
[253]	52AB5B00-08D0-3D09-A730-DF586F680E3D	0x0000000019adde000	/System/Library/PrivateFrameworks/CorePhoneNumbers.framework/Versions/A/
[254]	EC014994-D30C-3921-8B6E-506D56AA0F61	0x00000000195e85000	/usr/lib/libiconv.2.dylib
[255]	2D3D172D-7297-3AFE-B4E8-13C6BBF101B6	0x0000000019ac9f000	/System/Library/PrivateFrameworks/MobileAsset.framework/Versions/A/Mobi
[256]	04B9B3F9-B2B3-39E9-9A2A-FB5F34F61708	0x000000001cd2aa000	/usr/lib/libpartition2_dynamic.dylib
[257]	77297519-C230-3D29-8C97-6758EFE7F717	0x000000002aa5b4000	/usr/lib/libReverseProxyDevice.dylib
[258]	C0062537-08DF-37FE-B544-68F8953E1DA1	0x000000001ca27a000	/usr/lib/libamsupport.dylib
[259]	62668A02-83A4-3A60-B600-FAD888935521	0x0000000019a90a000	/System/Library/PrivateFrameworks/MultiverseSupport.framework/Versions/A/
[260]	279369FC-BDAA-3B3E-AE39-C8793C553A2F	0x000000001a4253000	/System/Library/PrivateFrameworks/RemoteServiceDiscovery.framework/Versi
[261]	191F441D-7E29-358E-973A-2898F82E6B9F	0x0000000019ada6000	/System/Library/PrivateFrameworks/KeychainCircle.framework/Versions/A/Ke
[262]	A6BBF231-9100-342E-9C20-6BFB5104529C	0x000000001a70e4000	/System/Library/PrivateFrameworks/Seeding.framework/Versions/A/Seeding

dyld_shared_cache

exec

exec

exec

exec

exec

exec

exec

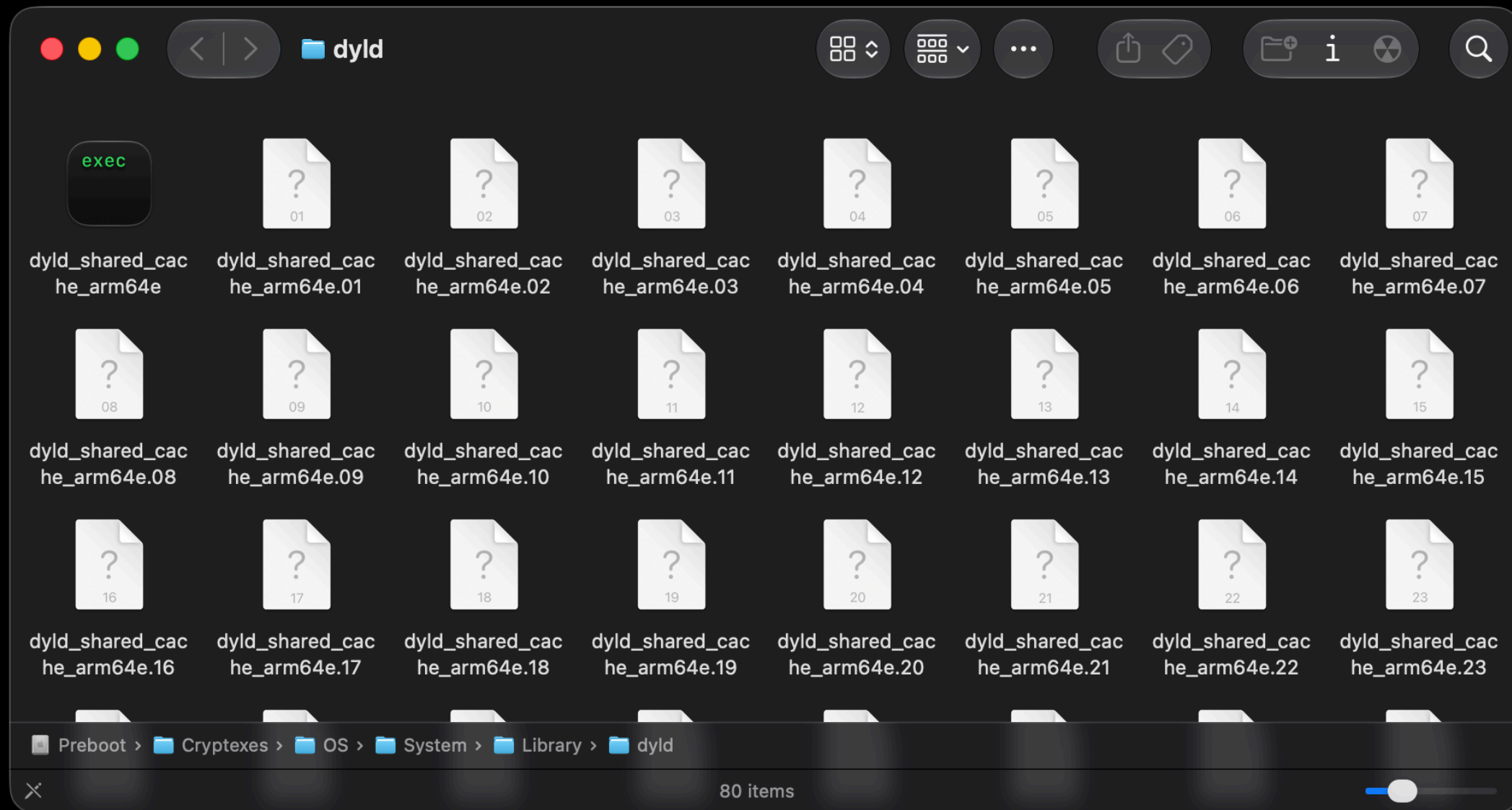
exec

dyld_shared_cache

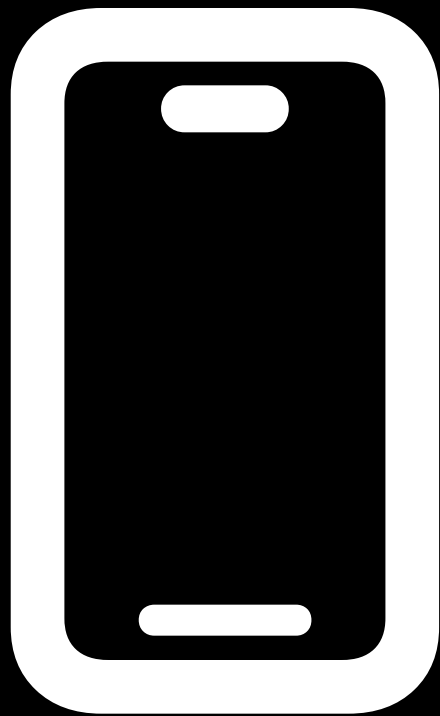


dyld_shared_cache

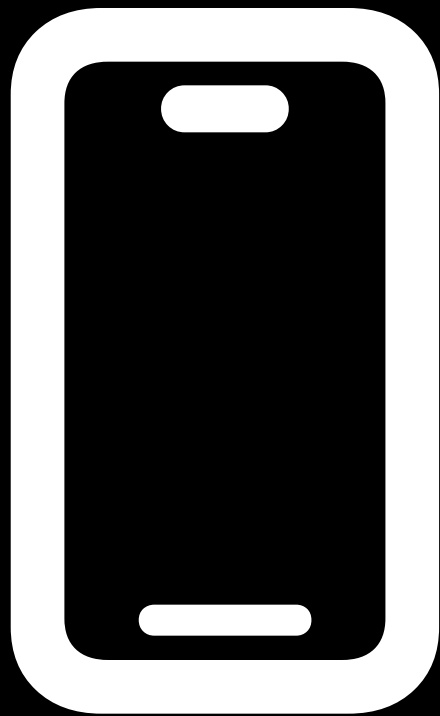
dyld_shared_cache



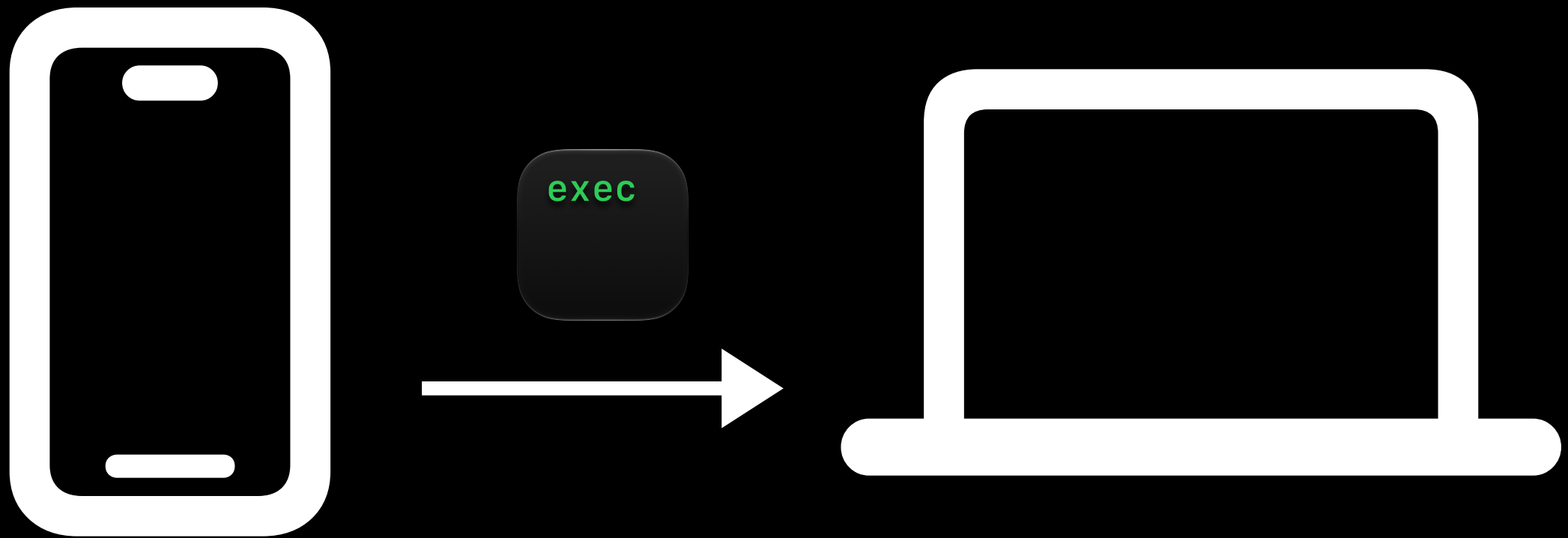
Device preparation



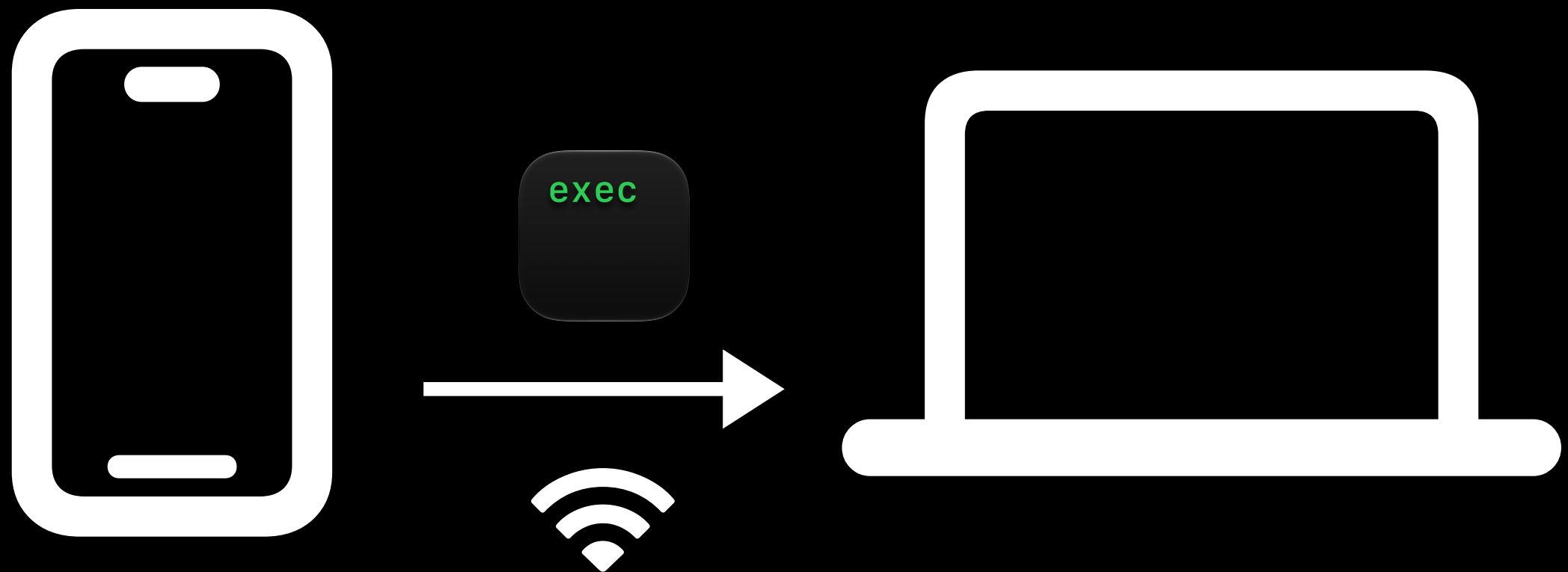
Device preparation



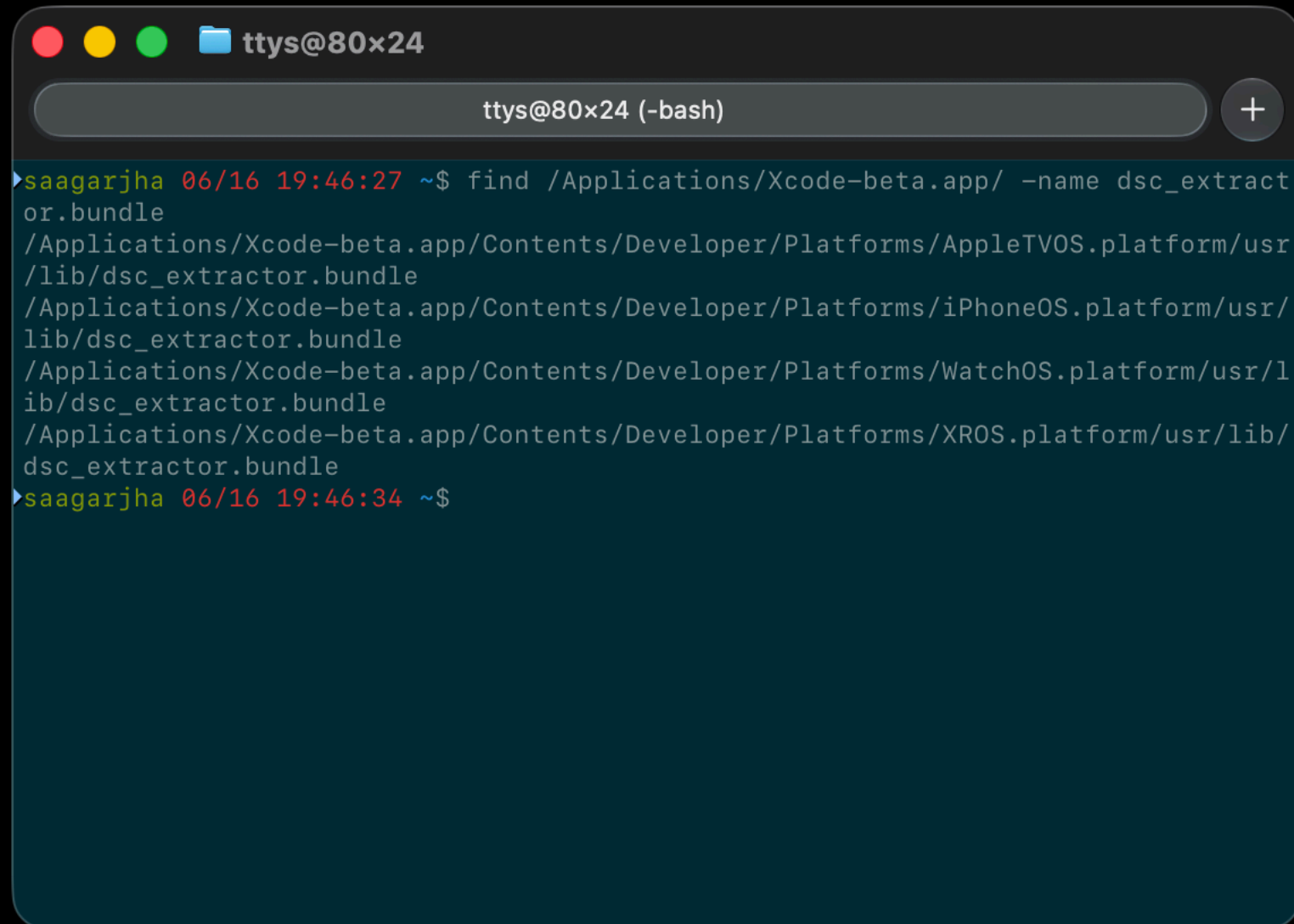
Device preparation



Device preparation

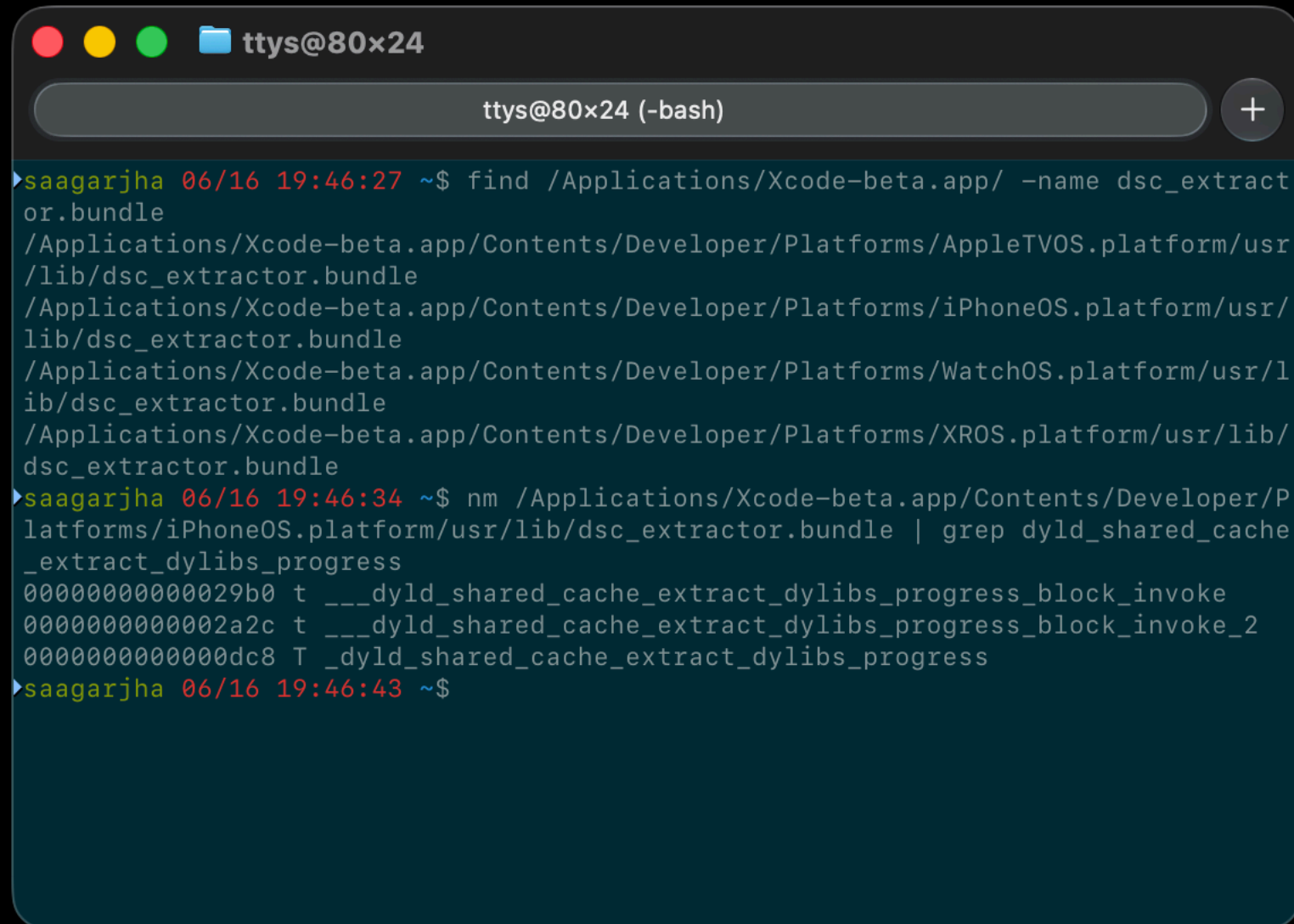


Device preparation



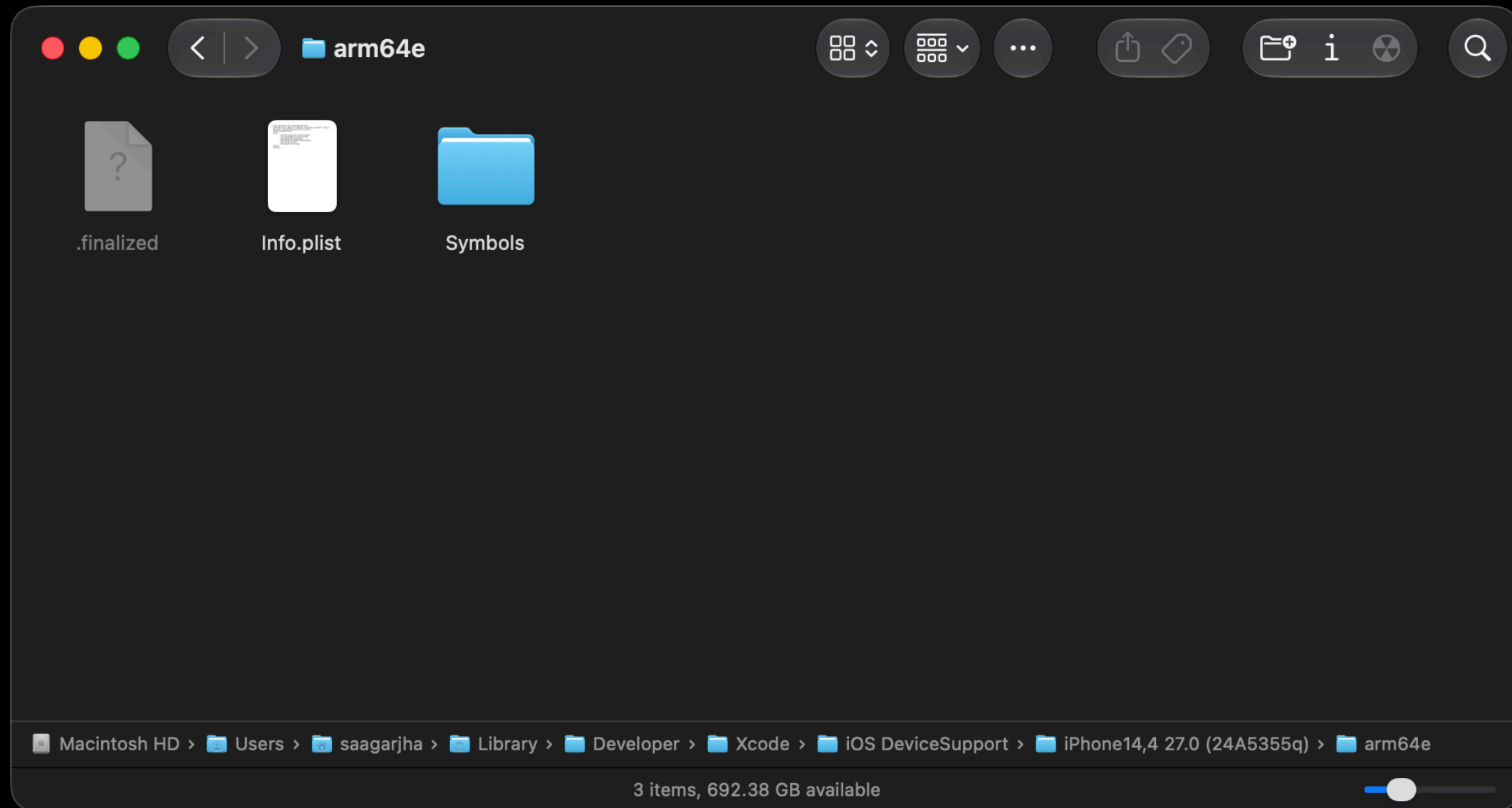
```
ttys@80x24  
ttys@80x24 (-bash)  
▶saagarjha 06/16 19:46:27 ~$ find /Applications/Xcode-beta.app/ -name dsc_extractor.bundle  
/Applications/Xcode-beta.app/Contents/Developer/Platforms/AppleTVOS.platform/usr/lib/dsc_extractor.bundle  
/Applications/Xcode-beta.app/Contents/Developer/Platforms/iPhoneOS.platform/usr/lib/dsc_extractor.bundle  
/Applications/Xcode-beta.app/Contents/Developer/Platforms/WatchOS.platform/usr/lib/dsc_extractor.bundle  
/Applications/Xcode-beta.app/Contents/Developer/Platforms/XROS.platform/usr/lib/dsc_extractor.bundle  
▶saagarjha 06/16 19:46:34 ~$
```

Device preparation

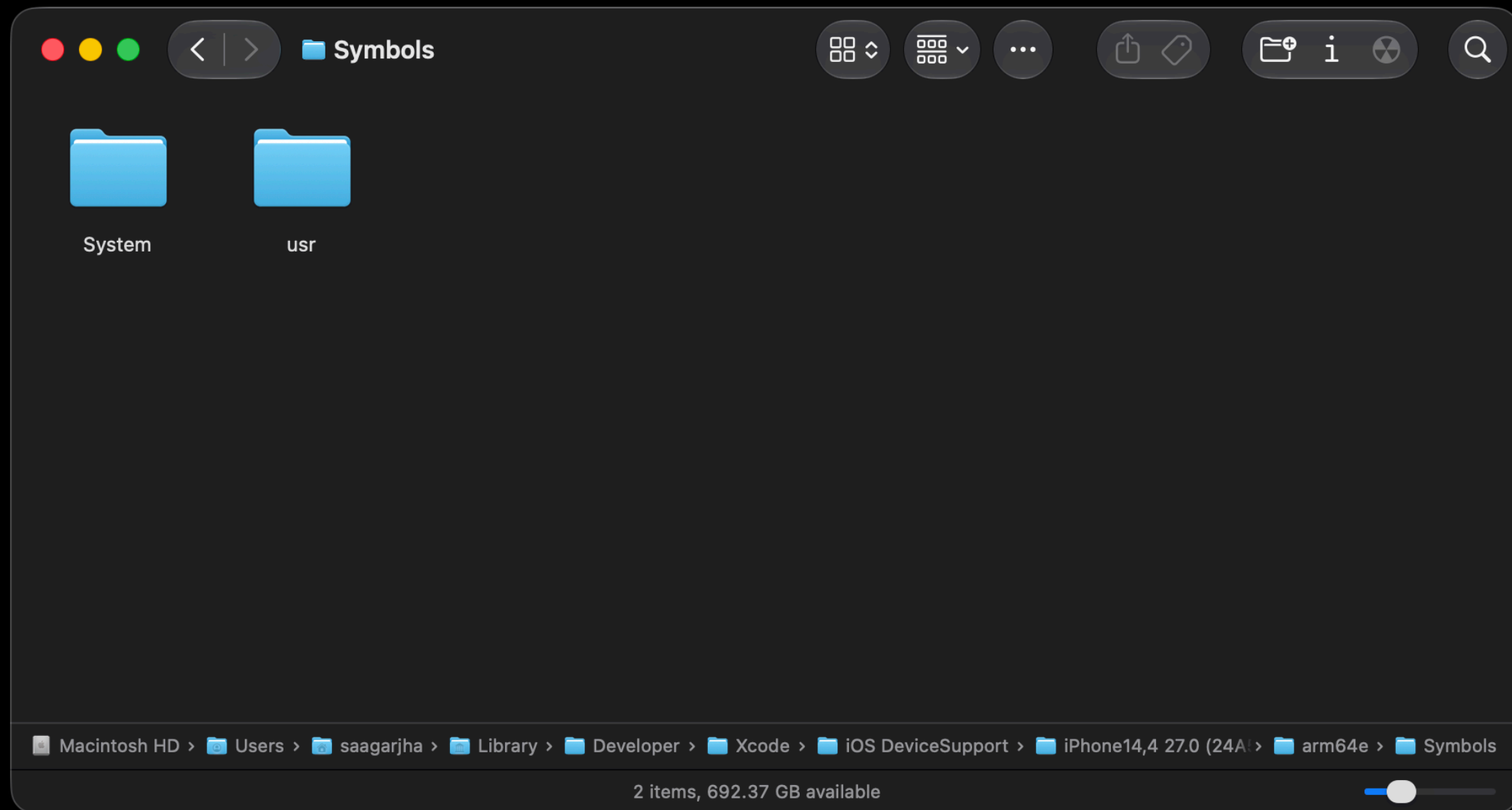


```
ttys@80x24  
ttys@80x24 (-bash)  
▶ saagarjha 06/16 19:46:27 ~$ find /Applications/Xcode-beta.app/ -name dsc_extractor.bundle  
/Applications/Xcode-beta.app/Contents/Developer/Platforms/AppleTVOS.platform/usr/lib/dsc_extractor.bundle  
/Applications/Xcode-beta.app/Contents/Developer/Platforms/iPhoneOS.platform/usr/lib/dsc_extractor.bundle  
/Applications/Xcode-beta.app/Contents/Developer/Platforms/WatchOS.platform/usr/lib/dsc_extractor.bundle  
/Applications/Xcode-beta.app/Contents/Developer/Platforms/XROS.platform/usr/lib/dsc_extractor.bundle  
▶ saagarjha 06/16 19:46:34 ~$ nm /Applications/Xcode-beta.app/Contents/Developer/Platforms/iPhoneOS.platform/usr/lib/dsc_extractor.bundle | grep dyld_shared_cache_extract_dylibs_progress  
00000000000029b0 t __dyld_shared_cache_extract_dylibs_progress_block_invoke  
0000000000002a2c t __dyld_shared_cache_extract_dylibs_progress_block_invoke_2  
0000000000000dc8 T _dyld_shared_cache_extract_dylibs_progress  
▶ saagarjha 06/16 19:46:43 ~$
```

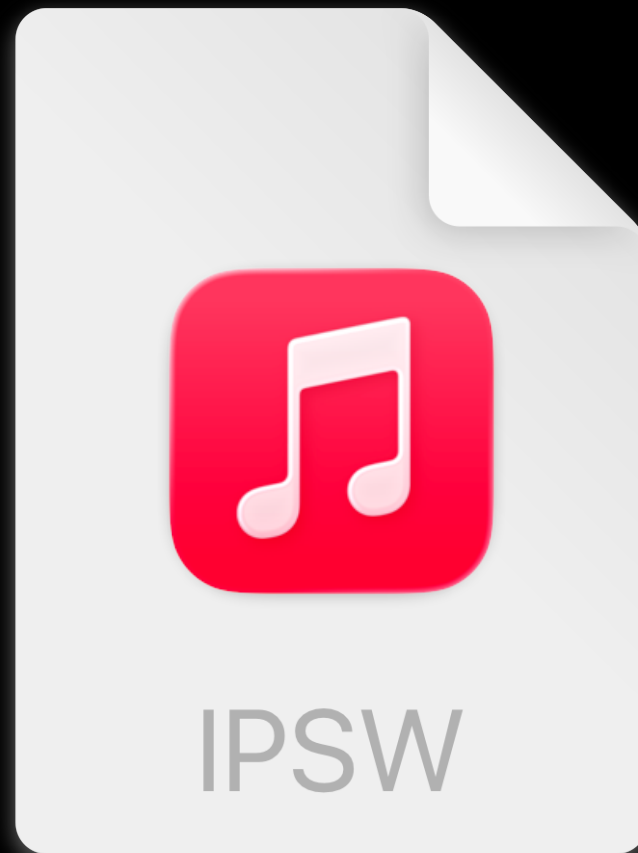
Device preparation



Device preparation



IPSW



IPSW



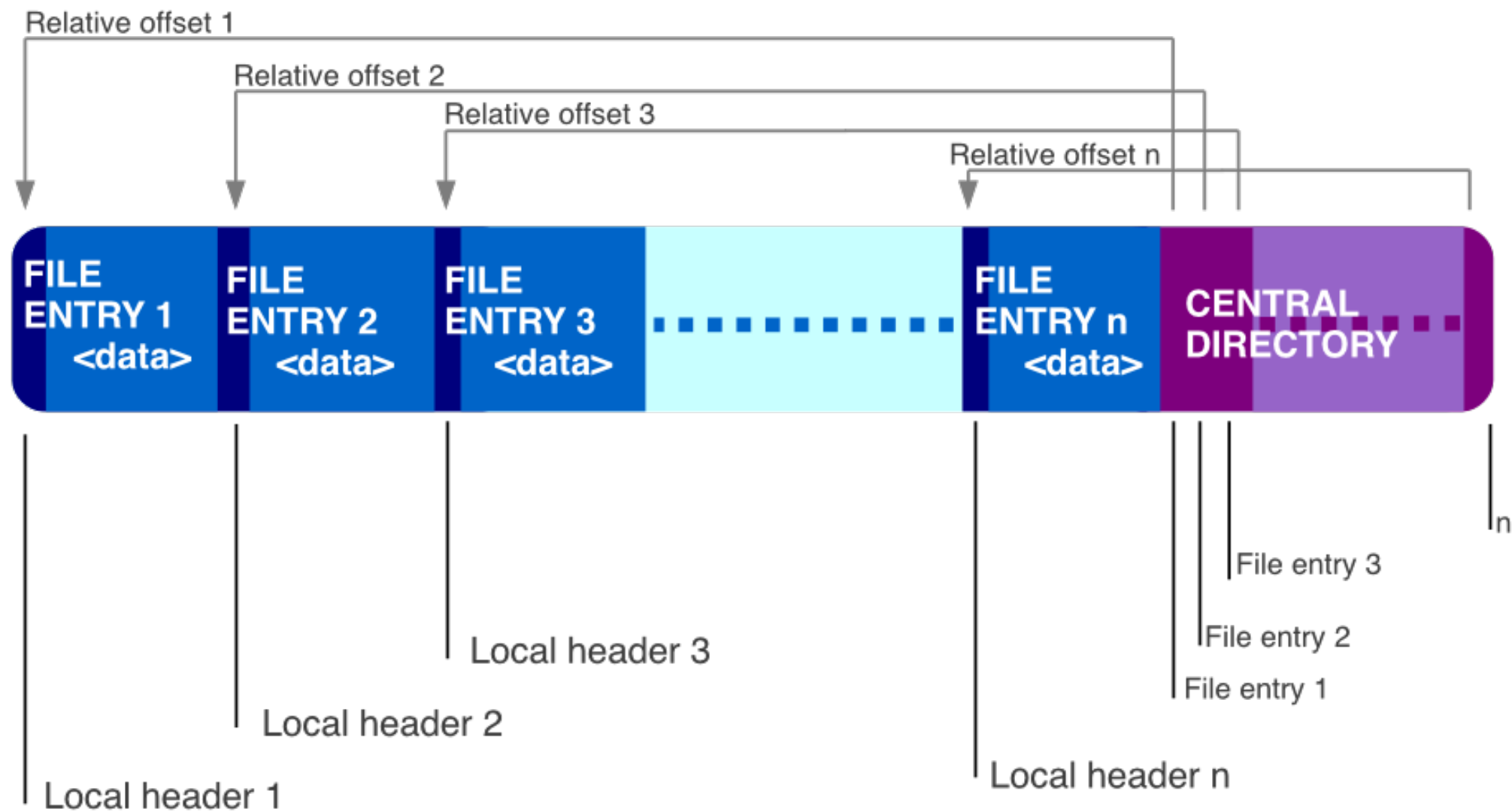
IPSW



IPSW



ZIP file format



ZIP file format

The screenshot shows a web browser window displaying the GitHub repository page for 'planetbeing/partial-zip'. The browser's address bar shows the URL 'github.com/planetbeing/partial-zip'. The GitHub navigation bar at the top includes links for Platform, Solutions, Resources, Open Source, Enterprise, and Pricing, along with a search bar and 'Sign in' and 'Sign up' buttons. The repository header shows the name 'planetbeing / partial-zip' with a 'Public' badge, and buttons for Notifications, Fork (25), and Star (69). Below the header, a navigation bar lists repository features: Code (selected), Issues (3), Pull requests (2), Actions, Projects, Wiki, Security and quality, and Insights. The main content area shows the repository's file structure and commit history. The commit history table lists the following files and their commit details:

File	Commit Message	Commit Date
include	Fix endianness fix to be universal (works on OSX)	16 years ago
CMakeLists.txt	Fixed utility.	10 years ago
README	Initial commit	17 years ago
main.c	Fixed utility.	10 years ago
partial.c	Fixed utility.	10 years ago

Below the commit history, the README file is displayed with the following content:

```
Use cmake 2.6 to create the Makefiles for this.
```

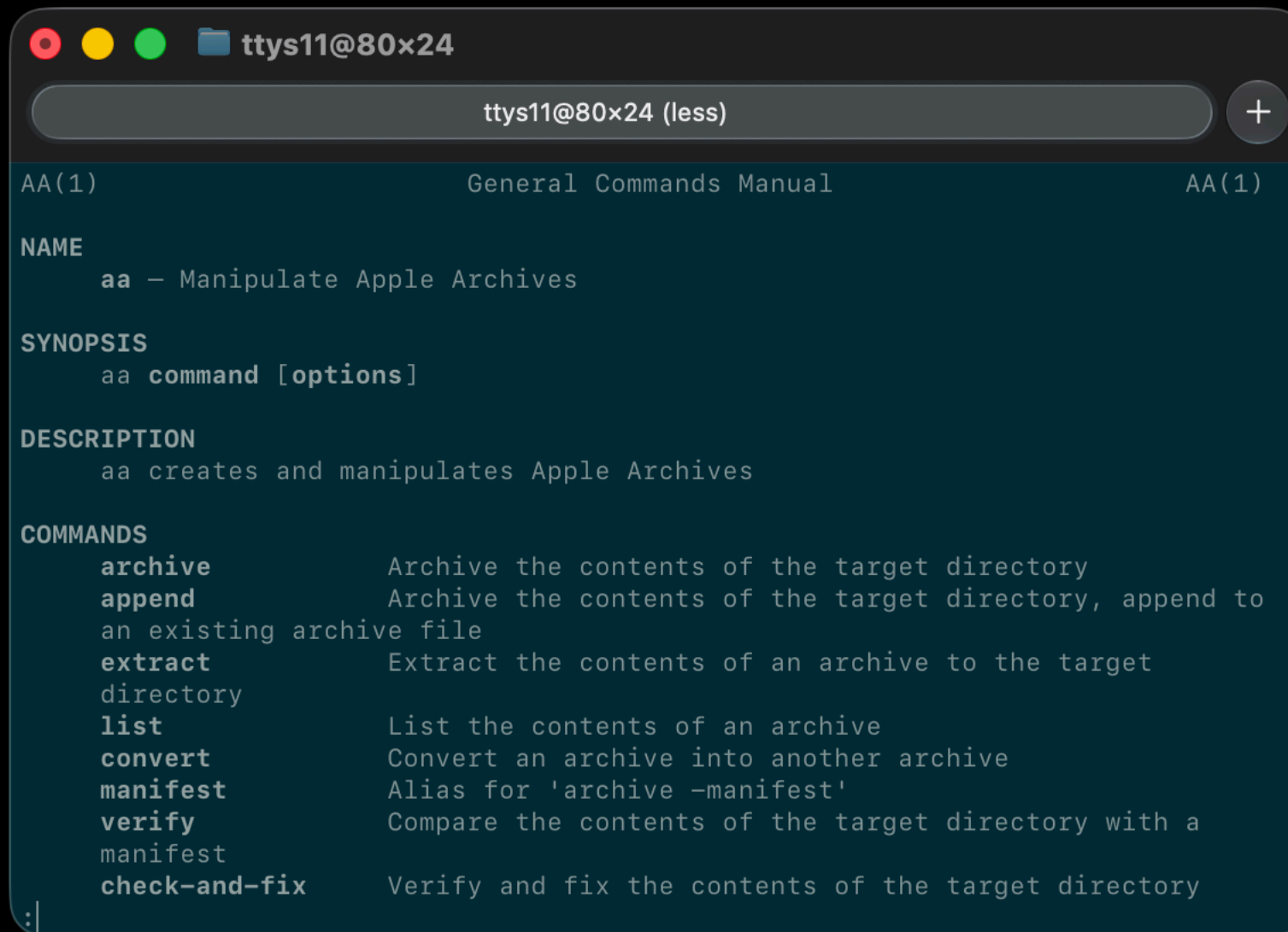
```
Requires libcurl 7.19.4 for file:// URL support.
```

The right sidebar contains the 'About' section, which describes the repository as a utility for retrieving selected portions of ZIP files using curl's partial downloading feature. It also lists the repository's statistics: 69 stars, 4 watchers, and 25 forks. Below the 'About' section, there are sections for 'Releases' (No releases published) and 'Packages' (No packages published).

Complications



Complications



```
ttys11@80x24 (less)
AA(1)                                General Commands Manual                                AA(1)

NAME
    aa - Manipulate Apple Archives

SYNOPSIS
    aa command [options]

DESCRIPTION
    aa creates and manipulates Apple Archives

COMMANDS
    archive           Archive the contents of the target directory
    append            Archive the contents of the target directory, append to
                        an existing archive file
    extract           Extract the contents of an archive to the target
                        directory
    list              List the contents of an archive
    convert          Convert an archive into another archive
    manifest         Alias for 'archive -manifest'
    verify           Compare the contents of the target directory with a
                        manifest
    check-and-fix     Verify and fix the contents of the target directory

:
```

Complications



Complications

The screenshot shows the GitHub interface for the repository 'saagarjha/unxip'. The browser address bar shows 'github.com/saagarjha/unxip'. The repository is public and has 956 stars, 46 forks, and 86 commits. The main branch is 'main'. The repository description is 'A fast Xcode unarchiver'. The file list shows various files and folders with their commit messages and dates. The right sidebar contains the 'About' section with links to the README, license, and activity, and the 'Releases' section showing the latest release '3.3' on Jan 8.

github.com/saagarjha/unxip

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saagarjha / unxip Public

Notifications Fork 46 Star 956

Code Issues Pull requests Actions Projects Security and quality Insights

main 1 Branch 13 Tags

Go to file

Code

saagarjha unxip 3.3 6c39905 · 5 months ago 86 Commits

.github/workflows	Run CI on newer OSes	5 months ago
Sources	Add a libunxip target	3 years ago
.gitignore	Add .swiftpm to .gitignore	4 years ago
.swift-format	Add a .swift-format config	4 years ago
LICENSE	Add a license	4 years ago
Makefile	Use swift build in Makefile	5 months ago
Package.swift	Support older Darwin OSes	5 months ago
README.md	Fix typo in README	last year
release.sh	Enable whole module optimization for binaries	3 years ago

About

A fast Xcode unarchiver

- Readme
- LGPL-3.0 license
- Activity
- 956 stars
- 19 watching
- 46 forks

Report repository

Releases 13

3.3 Latest on Jan 8

+ 12 releases

Packages

Demo

Future directions

- Deslop
- Device detection
- Automatic symbols cleanup
- Publish as open source

Questions?